ROUND 1 REVIEW

AI Frontier Challenge 2026 Round 1 Submission

TriageGuard: Clinical Decision Support System for Emergency Triage

Prototype on 30,000 synthetic encounters; prospective validation required before clinical use.

Team Name

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Team Leader

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Track

H1. OPD Triage Assistant for an Overloaded Government Hospital

Date

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Use Case

Problem Statement

• **Clinical Pain Point & Cognitive Overload:** High-volume government district casualty wards handle 800+ patients daily with <90 seconds per triage encounter. Severe cognitive fatigue leads to a documented 15–20% under-triage rate on subtle, life-threatening presentations (silent STEMI, occult sepsis, internal shock).

• **Multilingual & Demographic Communication Barriers:** Over 65% of patients in regional and district hospitals communicate in Telugu or Hindi, frequently describing life-critical symptoms through regional metaphors (*"gunde lo manta"*, *"dummu pattukundi"*) that standard medical NLP and non-native clinical staff misunderstand.

• **Quantified Patient & Systemic Impact:** Every 10-minute delay in emergency resuscitation increases patient mortality by 8–12%. Current hospital triage relies on paper tokens with 30% inter-nurse variability, while black-box LLMs hallucinate on corrupted vitals and get manipulated by patient queue-gaming keywords.

Market Opportunity

• **Market Size & Rapid Sector Growth:** The Global Clinical Decision Support System (CDSS) market stands at $2.1B (2025) and is projected to reach $4.8B by 2030 (18.2% CAGR). Digital emergency adoption is accelerated by India's Ayushman Bharat Digital Mission (ABDM) and public smart-hospital modernization mandates.

• **Target Audience & Hospital Scale:** Primary deployment targets high-footfall government district casualty wards (AP, Telangana, PAN-India public healthcare infrastructure) and tertiary trauma networks requiring auditable ESI Level 1–5 intake.

• **Competitive Advantage & Strategic Moat:** Manual ESI/Manchester triage is slow, paper-bound, and subjective; generic generative AI lacks clinical safety bounds. TriageGuard provides deterministic safety rules, calibrated Mondrian bounds, 12.8 ms mean, 16.2 ms P95 latency, and on-premise execution; no patient data leaves the hospital network.

Proposed Solution

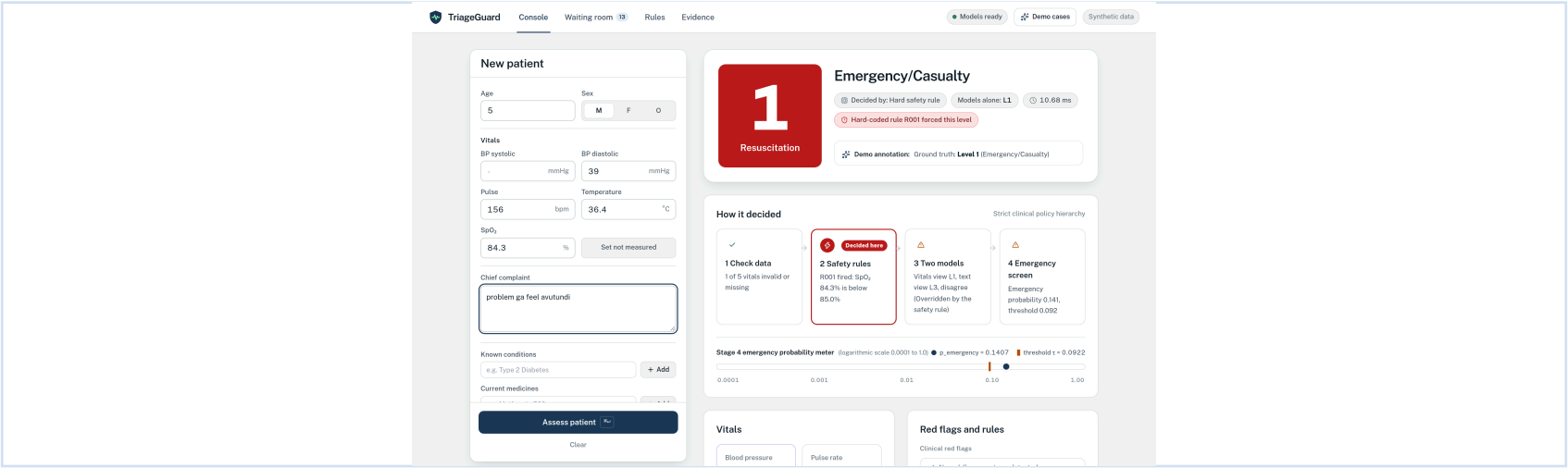
Solution Overview

• **TriageGuard CDSS Architecture:** Prototype on 30,000 synthetic encounters; prospective validation required before clinical use. Ultra-fast inference (12.8 ms mean, 16.2 ms P95) classifies ED acuity (ESI Levels 1–5) and routes to 12 specialties (88.5% on the 3,737 non-abstained cases).

• **5-Stage Safety Pipeline:** (1) Input Cleaning & Plausibility → (2) Deterministic Rules (13 deterministic thresholds, placeholder values pending clinical primer) & 10 Red Flags → (3) Monotonic LightGBM + Text Model (TF-IDF + Logistic Regression) → (4) Mondrian Conformal Screen & Safety Abstention Gate → (5) Cost Bayes Engine & 12-Dept Router.

• **Benchmark Metrics:** Emergency recall: 76.7% (138/180), vague-complaint stress, vs Baseline 68.9% (124/180); silent misses: 5 vs 11 on clean data (54%); 19 vs 56 under vague stress (66%); 4.4% non-emergency over-escalation (192/4,320); anti-gaming: 79.6% held at true level, 18.4% routed to nurse, 0.6% (3/500) reached L1/L2.

• **Safety Philosophy:** Rule-triggered cases are never downgraded. ML never holds a solitary life-or-death vote. Fails safely by abstaining 17% of clean cases (37.1% under 20% sensor corruption) to senior triage nurses.



Design Steps & Architecture

Technical Implementation

1. Multimodal Intake & Sensor Validation:
Continuous vitals (BP, SpO2, HR, Temp) + chief complaint text. Plausibility filters detect sensor faults; ≥2 invalid vitals trigger nurse remeasurement.

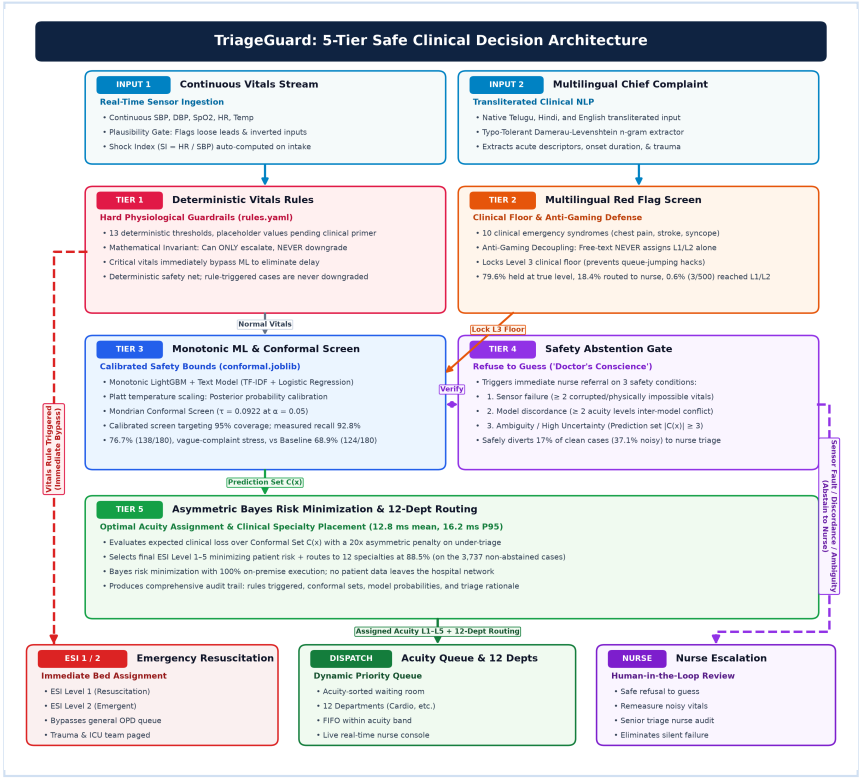
2. Rule & Red-Flag Floor Screen:
13 deterministic thresholds, placeholder values pending clinical primer, plus 10 red flags (Telugu, Hindi, English n-grams). Free text enforces Level 3 floor without allowing artificial Level 1/2 gaming (79.6% held at true level, 18.4% routed to nurse, 0.6% [3/500] reached L1/L2).

3. Monotonic LightGBM & Text Model:
Monotonic gradient boosting (SpO2 ↓, Shock Index ↑) fused with TF-IDF + logistic regression text model. Platt temperature scaling produces calibrated probabilities.

4. Mondrian Conformal Screen & Safety Gate:
Calibrated screen targeting 95% coverage; measured recall 92.8% (167/180, CI 88.0-95.7%). Safety gate abstains 17% of clean cases (37.1% under 20% corruption).

5. Asymmetric Bayes Risk & 12-Dept Router:
Minimizes expected risk with a 20x under-triage penalty inside prediction set C(x) and routes to 12 specialties at 88.5% (on the 3,737 non-abstained cases) in 12.8 ms mean, 16.2 ms P95.

Architecture Diagram



Challenges & Mitigation

Challenge 1

Problem: Patients inject alarming keywords (*"unbearable dying pain please see me now"*, *"chast pain emergency"*) at kiosks or intake desks to jump the triage waiting queue.

Mitigation: Architectural channel decoupling: free-text red flags **can never assign Level 1 or 2 by themselves**; they only enforce an acuity floor of Level 3.

Empirical Result: In 500 stress-tested gaming attacks: **79.6% held at true level, 18.4% routed to nurse, 0.6% (3/500) reached L1/L2**.

Challenge 2

Problem: Busy casualty wards suffer from detached pulse oximeter probes, inverted systolic/diastolic values, Fahrenheit entered in Celsius fields, or transient signal dropouts.

Mitigation: Deterministic physical plausibility gates. Rather than blindly guessing with statistical imputation, TriageGuard triggers `needs_remeasure` nurse intervention when ≥2 vitals fail.

Empirical Result: Abstention scales safely from **17% of clean cases** to **37.1% under 20% sensor corruption** while strictly preserving **92.2% emergency recall (166/180)**.

Challenge 3

Problem: Standard ML models optimize aggregate accuracy, silently missing rare crises (1–4% prevalence) such as atypical silent STEMI or occult respiratory failure.

Mitigation: Mondrian Conformal Safety Screen + 20x Asymmetric Cost Matrix. Calibrated on split emergencies (τ=0.0922, α=0.05).

Empirical Result: Emergency recall: **76.7% (138/180), vague-complaint stress, vs Baseline 68.9% (124/180)**. Silent misses: **5 vs 11 on clean data (54%); 19 vs 56 under vague stress (66%)**.

Unique Selling Proposition (USP)

Key Differentiator 1

Swiss-Cheese Safety Model

Deterministic Physiological Net

13 deterministic thresholds, placeholder values pending clinical primer, wrap the ML core. Invariant: Rules can ONLY escalate, NEVER downgrade. SpO2 < 85% or profound shock overrides statistical predictions; rule-triggered cases are never downgraded.

Key Differentiator 2

Mondrian Conformal Screen

Calibrated Statistical Bounds

Replaces heuristics with calibrated prediction sets C(x). Calibrated screen targeting 95% coverage; measured recall 92.8% (167/180, CI 88.0-95.7%) while limiting non-emergency over-escalation to 4.4% (192/4,320).

Key Differentiator 3

Honest Clinical Abstention

Doctor's Conscience Safety Gate

Clinical AI must know when NOT to guess. TriageGuard safely abstains under ≥2 corrupted vitals, inter-model discordance (≥2 levels), or high ambiguity, routing 17% of clean cases (37.1% under 20% sensor corruption) safely to senior triage nurses.

Competitive Advantage Summary

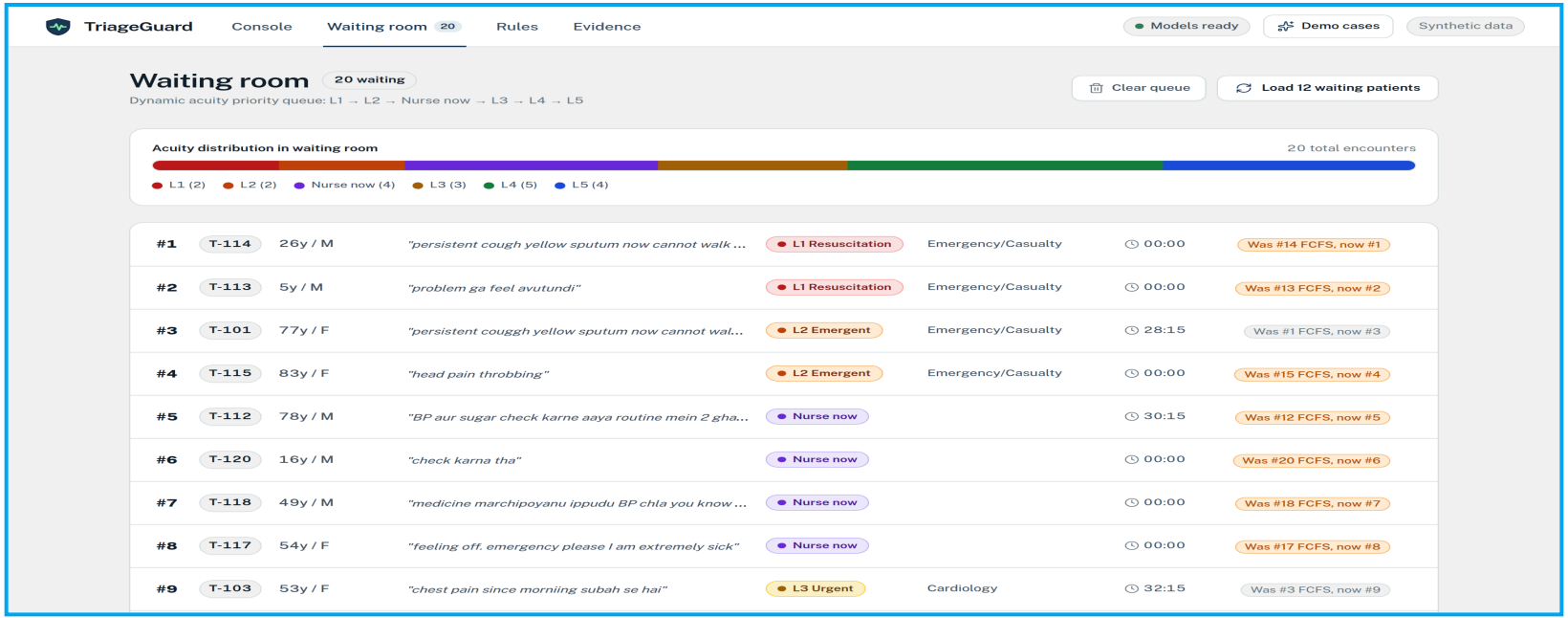
• **On-Premise Execution & Local Privacy:** Inference latency of 12.8 ms mean, 16.2 ms P95 enables 100% on-premise execution; no patient data leaves the hospital network, eliminating recurring cloud API costs.

• **Multimodal & Multilingual Synergy:** Fuses monotonic LightGBM with TF-IDF + logistic regression text model and typo-tolerant n-grams (Telugu, Hindi, English) to eliminate language and spelling barriers in district casualty wards.

• **Asymmetric Bayes Risk & 12-Dept Routing:** Evaluates expected loss with a 20x under-triage penalty; routes to 12 specialties at 88.5% (on the 3,737 non-abstained cases). Prototype on 30,000 synthetic encounters; prospective validation required before clinical use.

Demo & POC Video

Live Demonstration & Interactive Prototype System



[VIDEO] YouTube Demo

https://youtu.be/yqNZq92zks0

[CODE] GitHub Repo

https://github.com/grsanudeep42-cmd/triageguard

[DEPLOY] Live App

https://triageguard-ncai.onrender.com

[DOCS] Swagger API

https://triageguard-ncai.onrender.com/docs